

NDFEX Bot Mix

What you're trading against, and how it behaves

Trading competition debrief · Spring 2026

The two bot processes

Every fill on the exchange has a counterparty. With ~13 student teams active, the dominant counterparties are not other students — they are these two house bots.

bot_runner cid 99 · "test"

The **liquidity provider**. Always quotes both sides on every symbol. This is the "house market maker" you're scalping when you fire IOCs.

- 10 market-maker instances per symbol (7 generic + 3 dedicated UNDY)
- 1 stacker per symbol (inside-spread quote)
- 4–5 random-taker instances (low-rate noise flow)

smarter_bots cid 98 · "test2"

The **active participant**. Doesn't quote — it generates directional flow and trades against whatever liquidity is available.

- PressureTaker — directional bursts
- ImbalanceTaker — trend-following
- RestingOrders — passive off-touch limits
- RandomTaker — random side IOCs

Last session: cid 99 lost \$7.2M and cid 98 won \$6.7M. The system is designed for these two to push prices around — your job is to read what they're doing and trade accordingly.

bot_runner architecture

Every symbol gets the same stack of LP components. Each component has its own role in the BBO.

RandomWalkFairValue (per dorm/GOLD/BLUE)

The "true price" the LP is quoting around. Random walk that steps every 1–100 ms.

BasketFairValue (UNDY only)

Sum of component touches (10 dorm bids on the bid side, 10 dorm asks on the ask side). Reactive to component BBO changes.

FairValueMarketMaker × 7 generic + 3 UNDY

Quotes $\text{bid} = \text{FV} - W \cdot \text{tick}$, $\text{ask} = \text{FV} + W \cdot \text{tick}$, with per-MM variance offset and per-share position skew.

FairValueStackingMarketMaker (1 per symbol)

Posts one tighter "stacker" quote 1 tick inside the BBO. Provides inside liquidity that beats the generic MMs to the touch.

The BBO you actually see and trade against is usually the **stacker** on top of the generic MMs underneath. When the stacker gets hit, the generic MMs become the new BBO — usually 1 tick wider.

RandomWalkFairValue

The fair-value clock for every non-basket symbol. Drives the LP's quotes for GOLD, BLUE, and the 10 dorms independently.

How it moves

- Random walk: at each step, FV moves uniformly in `[-tick/5, +tick/5]`
- Step interval: uniform `[1ms, 100ms]` — so ~10–20 FV updates per second
- Bounded between `min_price + 5·tick` and `max_price - 5·tick`

Implications you can act on

- **FV updates are slow relative to your IOC.** The LP doesn't re-price on every event — only when its own internal tick fires.
- **FVs are independent across symbols.** KNAN and DILN walk separately. UNDY's basket FV is the only thing tying them together.
- **Bear-regime drift.** The instructor can fire a directional drift overlay that adds a deterministic linear move on top of the random walk for a configurable duration.

BasketFairValue (UNDY)

UNDY is structurally tied to its 10 dorm components. Its LP fair value is computed from the actual market touches, not a separate random walk.

How it computes

```
bid_value(t) = sum over 10 dorms of best_bid(dorm)
ask_value(t) = sum over 10 dorms of best_ask(dorm)
LP_UNDY_bid   = bid_value - W·tickUNDY
LP_UNDY_ask   = ask_value + W·tickUNDY
```

Why this is structurally arb-resistant

- By construction, $LP_UNDY_ask > \text{sum}(\text{component_bids})$. You cannot lift the LP's UNDY ask and immediately offset by selling the components — the cost exceeds the revenue by the LP spread.
- Same in the other direction: hitting LP_UNDY_bid and lifting the components loses money.

There is also a small (0.01% of calls) *pricing-error injection* that briefly returns mid-sum instead of touch-sum. The intent is to leak occasional opportunities; in practice the leak is small and rarely arbable on its own.

FairValueMarketMaker

The workhorse LP. 7 generic instances each quote every symbol, plus 3 dedicated UNDY instances. All share the same FV objects but apply per-MM offsets.

Quoting formula

```
bid_fv = FV.bid_value() + variance[mm][sym] - position
ask_fv = FV.ask_value() + variance[mm][sym] - position
bid     = bid_fv - W*tick      (then snapped to tick grid)
ask     = ask_fv + W*tick
```

Two key parameters

- **Variance vector.** Each MM has a different per-symbol offset, e.g. neutral / slightly bullish / contrarian. Different MMs end up at the BBO depending on which way the offsets line up.
- **Position skew.** When the MM has accumulated inventory, both bid and ask shift opposite to the position. Long → quotes drop. Short → quotes rise. Capped at `|position| ≤ 100`.

Position skew is the core mechanism that creates "momentum" in the LP's prices. Hitting one side repeatedly drives that MM short, which lifts both quotes — others can read this as a directional signal.

Stackers (FairValueStackingMarketMaker)

One stacker per symbol. Quotes **1 tick inside** the prevailing BBO with small size. Usually IS the BBO.

What this means in practice

- The price you see at top-of-book is usually the stacker, not the generic MM
- Stacker depth is small (typically 2 lots). Once you take the stacker, the next layer is the generic MM at $\pm W \cdot \text{tick}$ further out
- For UNDY only: the stacker uses the BasketFairValue to stay outside the structural arb range. For dorms, GOLD, and BLUE: the stacker is BBO-anchored (not basket-aware)

Why it matters

- Spread you observe $\neq$ spread you'd get for a large order — depth matters
- The stacker re-quotes when the underlying BBO moves, so it's slightly faster than the generic MMs to react

smarter_bots: PressureTaker

Generates directional flow. The single most impactful component for prices on short timescales.

Algorithm

1. Pick a random side (50% buy / 50% sell)
2. Pick a random total quantity (1..100)
3. Fire IOC's on that side every 200ms..10s
4. When total quantity is exhausted, GOTO 1

What you observe

- One-directional bursts of taker flow lasting seconds to a couple minutes
- Each burst pushes the LP MM into inventory → the LP's position skew amplifies the move
- When a burst exhausts, the next burst is independently random — half the time the direction reverses

This is the **main source of intraday "trend"** on the exchange. Reading order-flow from the trade tape and joining a burst early is a viable strategy. Exiting before the burst exhausts is the harder part.

smarter_bots: ImbalanceTaker

Watches trade-summary aggressor flow. Joins the dominant direction once it crosses a threshold.

Algorithm

```
maintain imbalance[sym] = sum of (signed) aggressor qty
on each trade summary:
    if aggressor was BUY: imbalance[sym] += qty
    else:                 imbalance[sym] -= qty
every 100ms:
    if imbalance[sym] > threshold[sym]: fire BUY IOC, reset
    if imbalance[sym] < -threshold[sym]: fire SELL IOC, reset
```

Effect on price action

- **Trend amplifier.** Once enough aggressive buying hits the tape, the ImbalanceTaker piles in on the buy side too — pushing the LP further short and accelerating the move
- Different threshold per symbol (more sensitive on illiquid names)
- Combined with PressureTaker, you get a feedback loop: PressureTaker creates the initial push, ImbalanceTaker amplifies it, LP skew carries it further

smarter_bots: RestingOrders & RandomTaker

RestingOrders

Posts passive limit orders 3 ticks off the touch on a randomly-chosen side, every 0.5–10s.

```
side = random()
if BUY: price = best_bid - 3*tick
if SELL: price = best_ask + 3*tick
```

These usually don't fill — they just add depth a few ticks back from BBO. When the market sweeps that level, you'll see large fills against cid 98.

Neither of these is "smart" — they exist to add background noise and the occasional large resting fill. The directional logic lives in PressureTaker and ImbalanceTaker.

RandomTaker

Bursts of small, random-side IOC takes against the touch. Several instances run with different time scales (10–500 ms base, scaled).

```
if random() < sell_bias: SELL   IOC at best_bid
else:                   BUY    IOC at best_ask
```

The instructor can flip `sell_bias` from 0.5 to a sell-heavy regime (~0.85) for the bear-regime event.

Per-symbol setup

Tick size, MM width, and starting fair value for each tradable symbol.

Symbol	Name	Tick	Generic MM width	UNDY-only MM widths	Stacker width
1	GOLD	10	2 / 3 ticks	—	1 tick inside BBO
2	BLUE	5	2 / 3 ticks	—	1 tick inside BBO
3–12	10 dorm components	5	1 / 2 ticks	—	1 tick inside BBO
13	UNDY (ETF of dorms 3–12)	10	1 / 2 ticks	1, 2, 3 ticks (3 layers)	1 tick inside BBO (basket-aware)

UNDY mechanics

- 1 UNDY $\equiv$ 1 share of each of the 10 dorms (KNAN, STED, FISH, DILN, SORN, RYAN, LYON, WLSH, LEWI, BDIN)
- ETF service exposes **create** (give 10 dorms, get 1 UNDY) and **redeem** (give 1 UNDY, get 10 dorms) at zero cash
- Conversions don't generate fills — they appear as position adjustments

Risk rules & how they trip

Position limit

- Per-symbol high-water-mark: **10 lots** (long or short)
- Once you ever exceed it, that symbol is permanently flagged for the session
- On the flagged symbol, any positive PnL is clamped to zero — you lose your gains there for the rest of the day
- **Includes ETF adjustments.** Redeeming UNDY adds to all 10 dorm positions in one event — manage your dorm inventory before redeeming

PnL drawdown freeze

- If your live total PnL drops below the drawdown threshold, your score is **frozen at that value** for the rest of the session
- The leaderboard shows `frozen_pnl (live_pnl)` — the parens are the would-have-been if the freeze hadn't fired
- Position-limit clamps feed into total PnL, so a single bad redeem can cascade into the freeze

The freeze and clamp are independent. You can hit either or both. Most freeze events this week have been triggered indirectly: position clamp zeros positive PnL on multiple symbols → total drops below threshold → freeze fires.

What's been working (and what hasn't)

Working

- **Per-symbol momentum scalping.** Read PressureTaker bursts off the trade tape, join early with 1-lot IOCs, exit before the burst exhausts. ~1–2 ticks of edge per round trip after fees
- **Inside-spread market making.** Quote tighter than the LP stacker, collect spread when burst flow runs over you in your favor
- **Disciplined risk management.** Tight stops on losing trips (≤ 3 ticks) and asymmetric position sizing

No longer working

- **Static UNDY-vs-basket arb against LP.** The basket-aware UNDY MM removed the structural edge. Component widths were also tightened
- **Big ETF conversion bursts.** Without rebalancing in between, redemptions push orders through the position limit and trip the freeze
- **Holding directional inventory through volatility.** The drawdown freeze fires on transient MTM swings, not just realized losses

Takeaways for tomorrow

- **The exchange is mostly bots talking to bots.** ~80% of typical fills are against cid 99 (LP) or cid 98 (smarter_bots). Your edge comes from reading what they're doing
- **There is no explicit mean reversion in any bot.** The reversion you observe is emergent: PressureTaker bursts cycle direction at random, and the LP's position-skew cap limits one-way pushes
- **Order flow is your signal.** The trade tape — who's hitting which side, how aggressively — encodes everything that matters about the next few seconds of price action
- **Risk rules are unforgiving.** One bad redeem can wipe an entire session of legitimate trading PnL. Sequence your conversions to keep dorm positions $\leq 10 - \text{redeem_size}$
- **Tomorrow:** we'll go through specific tape examples and let you replay your own bots against this mix